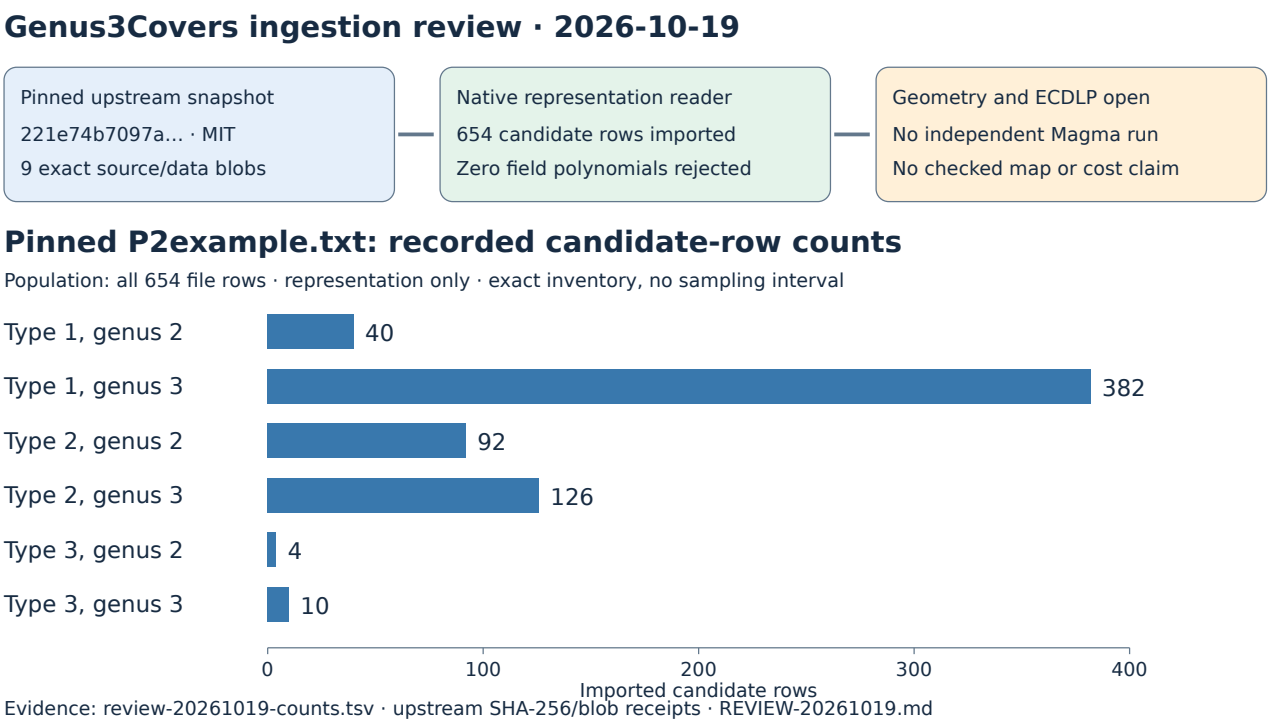
Genus3Covers: ingestion review

2026-10-19 · crypto PR 2241 · representation and provenance only

The immutable upstream revision is AndrewVSutherland/Genus3Covers, revision `221e74b7097a386527d3ebed68aa181b68954fd0`. All nine vendored/decompressed files match its exact Git blobs. The MIT license and the original Magma/example/catalog bytes are preserved.



Reader correction

The old native reader accepted [0,0] as a degree-one field polynomial. It now rejects zero leading coefficients, including negative-zero spellings and padded high-degree zeros. Nonzero arbitrary-precision coefficient strings are retained. The native importer tests pass on all 654 example rows and the adversarial representation controls; malformed input exits 2 without emitting partial JSONL. This is a degree/shape check, not an irreducibility or geometry certificate.

Research boundary and graph impact

These exact counts describe candidate rows in a pinned file, not verified covers or sampled performance. Magma is unavailable and was not run. No finite-field model/map, subgroup transfer, ECDLP cost, timing or speedup is promoted. Canonical covers/browser/scoreboard/timeline and benchmark claims are unaffected; adding such claims still needs their native proof and admission gates. The required full release library suite passed: 4,396 passed, zero failed, 118 existing ignored, no filtering. Final-head GitHub CI remains pending.

Sources, metadata TSV, upstream SHA-256/blob receipts, editable figure and reproduction commands are beside this PDF in REVIEW-20261019.md and the original ingestion README.